
Section 1 - Project Summary

Please describe your activities and any delivered or expected impacts of these actions at your host institution during your award. The following details would be particularly useful:

- The approximate number of students and staff you have engaged with
- If you have taught a formal course, include brief details
- The number of students you have helped obtain industry placements, short or long term positions
- The number of early stage companies you have mentored or supported in your University, please give brief details including level of involvement and current number of full-time employees
- The level of follow-on funding resourced with your support, please give brief details including breakdown between government, NGO, and private sector funding

Over the duration of the EiR placement at the University of Southampton I have undertaken the following activities:

One-to-one meetings and individual careers advice has been available to any and all postgraduate students and researchers. This has been taken up by ~20 individuals within the School of Physics and Astronomy with additional conversations being had with many others at events organised by the technology transfer office (RIS) and the university startup accelerator (Future Worlds). In total 50-100 individuals have been provided with some level of guidance or advice on the workings of commercial data science and its relevance to academia. Six have transitioned from academic into data science or related roles; several more have expressed to do so once they complete their studies. Many of those who have decided to stay within academia are adopting machine-learning and data science tools to enhance their research methods.

Networking and general data science career information has been delivered:

- I accompanied a number of PhD students to a variety of PyData events including:

PyData London conference 2018

PyData London conference 2019

Monthly PyData meetups, these were advertised to the students and they were free to attend if they were interested

- I have joined the Southampton Future Worlds Team that stimulates students and staff to start their own companies. I regularly attend their events to offer insight and guidance. This has also included giving more specific, detailed advice where appropriate. For example Dr Joshua Steer asked for 1-to-1 time to discuss the relevance of data science to his new company Radii Devices and advice on how to hire a data science team.

- I invited Pivigo (a data science training company) to come and give a cross-department seminar on data science career development and transitioning from academia. For those unable to attend I advertised subsequent webinars where more information was shared.

- I formed an organising committee to launch PyData Southampton, a local chapter of the international python data science community. We have a core-group of 6 volunteers and have sourced a commercial sponsor to fund the monthly meetup hosting and catering costs. Unfortunately due to the COVID19 pandemic we had to indefinitely postpone the launch of this new community group.

Technical skills training and development has included:

- I have arranged an annual git version control training workshop. This has run twice with a total of approximately 40 academics attending. Attendees ranged from PhD students to professors.

- I have organised an introductory workshop using JupyterLab as a data science workbench; this has

included one-on-one sessions to demonstrate some of the more advanced use cases.

- I have given a general seminar on the use of Docker for containerisation and delivered one-on-one instruction and training in how to use this tool to run encapsulated software.
- I have partnered with EIR James Otter to deliver an 8-part Commercial Literacy seminar series to academics across the university. This course has been run twice in-person; once on the Highfield Campus and once at Southampton General Hospital. The course materials are currently the basis for a virtual program being delivered to the Royal Society Industrial Fellows and others.
- I have been co-supervising several PhD students who are incorporating data science techniques into their thesis work including ML and GPU programming. My company, HAL24K, hosted one of these PhD students for a 3-month work placement in 2019.

Data4Good Social Impact activities:

- I arranged the 1st DataKind UK DataDive in Southampton in September 2018. It included 81 participants including 32 academics from within the university and 35 analysts from the Office of National Statistics and other government agencies. The two charities supported were Shelter and Parkinson's UK.
- In August 2019 I organised a Meetup in Southampton hosted by Snowflake Software to advertise the Data4Good activities we were doing to the wider business community to trigger interest in our next DataDive. Approximately 30 people attended.
- I arranged a 2nd DataKind UK DataDive in Southampton in September 2019. On this occasion there were 50-60 participants, 15-20 of which were from within the University and the other volunteers from the private sector companies and government agencies in and around Southampton. The charities supported were Meningitis Now and Stop the Traffik.
- I mentored and escorted PhD students and postdocs to 3 London based DataKind UK DataDive events; a total of 6 volunteer slots were taken by 5 academics from Southampton and worked on a variety of projects.
- The Data4Good activities were recognised by the University of Southampton Vice Chancellor Awards in 2019 and were shortlisted to receive the award for Research Impact.
- The School of Physics and Astronomy at Southampton has now completely bought into the concept of social impact data science projects and they see the value it brings to their staff and students. Consequently they intend to continue and build on these activities and we are in discussions as to how they can do more.

SPRINT Steering Board membership; the Space Research and Innovation Network for Technology, Southampton group asked me to join their steering board in October 2018. The University of Southampton is one of five UK universities which access to a pot of approximately £4.2 million to enable knowledge transfer from the university to SMEs.

- I have regularly attended the monthly steering board meetings for the past 18 months and contributed to planning and events.
- I attended the launch workshop where we facilitated introducing various academics to the SMEs.
- I have engaged with multiple academics within the School of Physics & Astronomy to advertise the SPRINT funding; as a result two collaborative tech transfer projects were established with two local SMEs (Symetrica and vivaMOS).

During my tenure as a Royal Society Entrepreneur in Residence I have been invited to speak at a number of conferences and events in which I have had the opportunity to advertise my EIR position and my general belief that there is more that can be done at the intersection of commercial data science and academia. This has facilitated me sharing the general EIR message beyond the University of Southampton and has included presentations and engagement at:

- PyData London, April 2018
- Erasmus Digital Economy Summit, Rotterdam, June 2018
- Leadership Challenges with Big Data, Rotterdam, January 2019
- Mathematics for Social Activism, Leeds, June 2019

- Erasmus Data Analytics Summit, Rotterdam, July 2019
- Human Centered AI, EGG Amsterdam, November 2019
- Next Generation Computational Modelling Group Guest Seminar, Southampton, February 2020

Funding has been arranged to support a number of the activities:

Data4Good activities have received:

- £500 from the Ordnance Survey
- £21,000 from the University of Southampton STFC Impact Acceleration Award
- £15,000 from the University of Southampton Astronomy Group DISCnet Impact grant
- Venue and hosting of Southampton DataDives on University of Southampton premises at no cost including car parking and wifi access

PyData Southampton activities have received or are promised:

- £500 from the Ordnance Survey
- Monthly meetup venue and catering from Ometria
- Central support from NumFocus which includes our Meetup account etc.

Which Impact are you most proud of?

I am most proud of building a new Data4Good community in Southampton that is rooted within the University of Southampton School of Physics and Astronomy because:

1. It is a non-traditional use of university staff and student time and yet I have managed to quickly get buy-in from the department leadership including the Head of Department and the Head of Astronomy. This is such that they have been prepared to give workload credit to staff participating in DataKind and Data4Good related activities. Additionally, they have sourced funds to support these activities.
2. There has been an incredible response within the academic community to participate in the events. This has resulted in both training and networking opportunities for all participants that is breaking the researchers out of their academic bubbles to see the value and impact their skills can have in a wider context.
3. We have been able to support more than 6 charitable organisations by application of academic analytical skills. This has delivered incredible value into these organisations and is directly delivering social impact and innovation.
4. The commitment to these activities has started conversations about how to make these engagements sustainable for the future beyond the end of the EiR position. In other words the organisation wants to continue doing this "new activity" and continue to develop how they can deliver social impact.
5. The value of these activities was recognised by the wider university through being nominated and subsequently shortlisted for a Vice Chancellor's Research Impact Award in 2019.

DataKind UK has been discussing the challenge of scaling their events and by partnering with Southampton we are exploring if university campus environments

Which year was the most impactful and why?

Both years were similar in impact. This may be attributed to the fact that I had a long-standing relationship with the department I was joining and so knew many of the parties involved. Also, the partnerships I had planned to bring into the university (e.g. DataKind) I had already established.

I was also fortunate in that the social impact work with DataKind, which I anticipated to be one of the

harder things to get started, accelerated very quickly in year one due to unexpectedly finding a funding source in year one and partnering with the Office of National Statistics.

However I have noticed an increased awareness of my role from people beyond the School of Physics and Astronomy in year 2. Clearly, it has taken time for "word to get around"; for example my first meeting with the Deputy Vice Chancellor for Research and Innovation occurred in April 2020 after the end of the initial EiR timeframe.

I suspect that if I arrived with a cold-start, i.e. no pre-established relationships, it may have taken a lot longer to deliver impact.

Please describe any problems that you have encountered during your Entrepreneur in Residence and how you have overcome them.

There have been no real problems that I have personally encountered except for when initially joining the University I had to negotiate with the Research and Innovation Services (RIS) for an appropriate "contract". I was originally given a 20-page document that required me to get insurance for millions of pounds to indemnify myself. I was able to renegotiate a more sensible agreement but it might be helpful if there was a boiler-plate agreement agreed between the RIS and the universities in question rather than needing to rely on the EiRs.

I have been fortunate that the university has made funding available to support some of the larger activities I had proposed without that support I would not have been able to host two local DataDives without finding £20-30k of commercial sponsorship.

Section 2 - Impact for the University

Has your host University's appreciation of industry and commercial links improved since the start of your award? What was your involvement in this change?

Over the course of the past two years I have interacted with different parts of the university in my role as an EiR; some parts of the university appear to have engaged more in an appreciation of industry and commercial links than others. With some groups I have seen a significant change but with others they seem somewhat indifferent.

The majority of my interactions and contact have been within the School of Physics and Astronomy. I'm happy to report that from my perspective they have shown a significant engagement in both the activities I have instigated as well as initiatives of their own. How much the appreciation of industry links, and entrepreneurial and innovation activities has been driven by myself and how much by requirements of REF 2021 are difficult for me to identify. Specifically the leadership of Physics & Astronomy have:

- Supported my "Data for Good" activities both financially and practically. They agree with the value of such activities and are building a relationship with DataKind UK which will last after the end of my EiR posting. They are actively looking to build social impact activities into their staff workload model and are preparing a business case to expand the scale and range of activities that they engage in under this umbrella.
- They have leveraged my industrial and commercial data science knowledge to deliver skills training across the postdoctoral researchers and have brought me in to advise on some PhD training.
- In part with my support and engagement with companies that have significant data science involvement I have seen more PhD students being offered the chance to suspend their PhD studies to do short industrial internships to get practical commercial experience. To my knowledge this is quite rare and is innovative within blue skies PhD research such as astrophysics.
- I have witnessed discussions in which they are thinking about how they can continue to access knowledge and experience equivalent to my own after the EiR ends.

My direct involvement with Research and Innovation Services (RIS) has been primarily limited to serving on the SPRINT Steering board. While there I have seen a desire to build more and stronger industrial and commercial links; within the scope of SPRINT it has been very successful. However, these initiatives are heavily dependent on external funding. As much as RIS may want to support more activities and to innovate they are constrained by their budgets and resources allocated from the University of Southampton leadership.

Recently myself and the other 3 Royal Society EIRs that have been appointed at the University of Southampton were invited to meet with the Vice Chancellor for Research and Innovation. He was very supportive of the activities that we explained we were undertaken but seemed to be rather happy with "where Southampton ranks in innovation and entrepreneurial activities" within the UK universities. It didn't feel like he felt there was a need to significantly enhance and improve in this area; although we did only have a single meeting.

My overall feeling from my experiences is that there is a lot going on a grass roots level of the university i.e. individuals and departments are engaging and innovating, and many of them really want to do things better and grow industrial and commercial links. However, there doesn't seem to be significant engagement at the level of the leadership of the University and hence there is limited support and budget that can filter down to the people who want to engage in these activities.

Please describe the level of support from the host institution. Is there anything they could've done to increase the impact of the scheme on their institution?

The support I have received from my host, the School of Physics and Astronomy, has been excellent. They have listened to my plans and supported my initiatives with resources, people and in some cases finances. They have both publicised my activities and have thought to invite me in to discussions where they think I can help them to build new links between industry and other university departments.

Through my engagements with RIS they have been supportive of my activities and have offered to assist if I identify activities in which I need their support. The central university has interacted at a much lower level and hasn't directly engaged with any of the EIR activities at an institutional level.

In my opinion the strength of the EIR programme is the independence of the people in post (by virtue of the fact that the EIR finances DO NOT go via the university). Consequently this facilitates the EIRs speaking "truth to power" as it were and engaging in independent activities. The value comes from the EIRs working at a grass roots level and so long as there is support and engagement at the department level they can do great things. Engaging on a strategic level with the University leadership is a very different type of activity and would require the leadership to recognise and want to improve that aspect of their business model. That being said if the message is carried through the grass roots it may in time percolate up to the leadership, but this may be a very slow process.

Section 3 - Benefit of the award

Please could you indicate if the grant has had or will have any positive impacts on your own career?

At age 40 I believe I am one of the youngest Entrepreneurs in Residence and as a result I believe that the grant has already had and will have a positive impact on my career. As a former academic I was aware of the RS but had never really engaged with them. Since becoming an EIR I have engaged in a number of workshops including one for the "Dynamics of Data Science Skills" report; I was rather proud that a significant number of things that I contributed in that workshop ultimately ended up in the report.

The RS brand is very powerful and it has given me additional credibility when I have been presenting at a variety of different conferences and meetings ranging from technical programming meetups through to data science ethics and society conferences.

The community of RS EiRs has also built my own professional network and given me new contacts and experiences that have changed how I will engage with future opportunities. Applying for the RS EiR grant has been one of the best decisions I have made in recent years.

Would you recommend the scheme to a peer or colleague and why?

Absolutely. I have been recommending it publicly for much of the past two years; in fact one of the new EiRs at Southampton applied as a direct consequence of my engagements and interactions in previous years.

My reasons can be summarised as.

- A new network that crosses scientific disciplines and industries.
- Access to the experience, advice and support of the Royal Society.
- An opportunity to influence and change how universities engage with industry with a good level of independence.
- An opportunity to give something back with the ability to commit time because of financial support i.e. the fact that it is paid and not simply volunteering time.

If there is anything you haven't covered and would like to add, please do so here.

On a personal level I hope that my engagement with the RS won't end with the EiR. I hope that the community and skills resource will be maintained in some way. If we can continue to engage in some of the workshops or future report working groups I think we can collectively add value. I certainly will be doing my best to stay in touch with those EiRs I have directly worked and partnered with.

I also wonder if the RS can support innovation and entrepreneurial activities at universities by encouraging them to work together on some of their approaches rather than purely competing. Whether this can be done through the pooling of resources in some manner I'm not certain but part of the problem that many universities appear to be experiencing is scaling the innovation activities with limited resources and budget.

Section 4 - Statement of Expenditure

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No Response